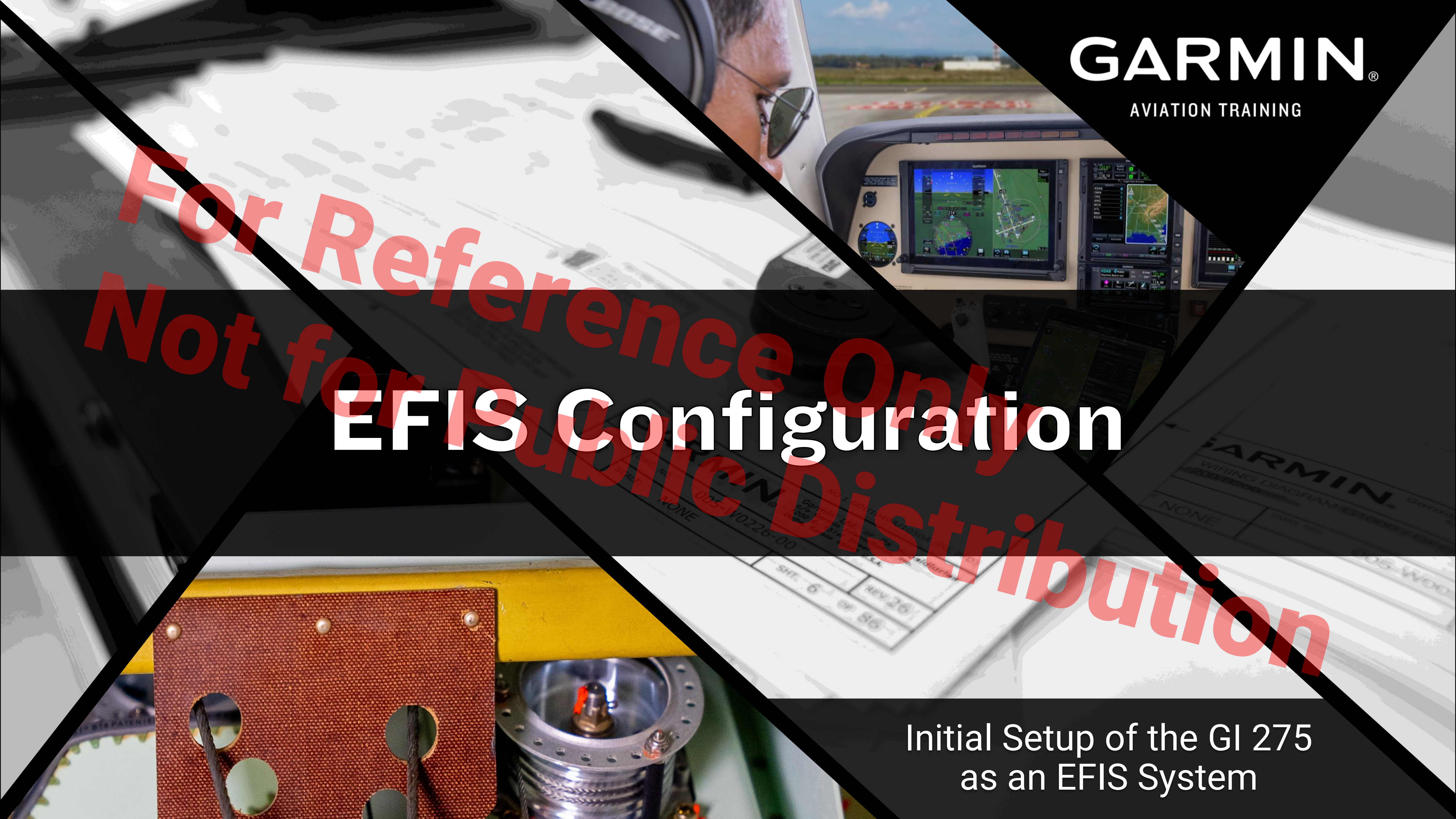


GARMIN®

AVIATION TRAINING

EFIS Configuration

Initial Setup of the GI 275 as an EFIS System

Legal Statement

Garmin Aviation Training is intended to provide the best practices for the installation, use and maintenance of Garmin Aviation products. It is incumbent on technicians and pilots to ensure that the instructions contained in the most recent versions of Garmin product manuals, bulletins and notifications are followed.

Objectives

Monday, March 30, 2026

- Focus on GI 275
  - Rules for being an ‘Electronic Flight Information System’ (EFIS)
  - Initial Setup and Configuration workflow
    - Identify
    - Configure
    - Calibrate

Objectives

Monday, March 30, 2026

- After completing this section, you will:
  - Understand the breakdown of the FAA's regulation for EFIS systems and how a network of GI-275's can offer 'Full Authority'
  - Identify work processes for better installation experiences.

14 CFR 23.1311 Electronic Display Instrument

- Paragraph B of the Subsection: The electronic display indicators, including their systems and installations, and considering other airplane systems, must be designed so that one display of information essential for continued safe flight and landing will remain available to the crew, without need for immediate action by any pilot for continued safe operation, after any single failure or probable combination of failures.

14 CFR 23.1311 Electronic Display Instrument

- Paragraph C of the Subsection: As used in this section, “instrument” includes devices that are physically contained in one unit, and devices that are composed of two or more physically separate units or components connected together (such as a remote indicating gyroscopic direction indicator that includes a magnetic sensing element, a gyroscopic unit, an amplifier, and an indicator connected together). As used in this section, “primary” display refers to the display of a parameter that is located in the instrument panel such that the pilot looks at it first when wanting to view that parameter.

14 CFR 23. 2615 Flight, navigation, and powerplant instruments.

- Paragraph A of Subsection: Installed systems must provide the flightcrew member who sets or monitors parameters for the flight, navigation, and powerplant, the information necessary to do so during each phase of flight. This information must:
  - (1) Be presented in a manner that the crewmember can monitor the parameter and determine trends, as needed, to operate the airplane; and
  - (2) Include limitations, unless the limitation cannot be exceeded in all intended operations.

- Paragraph B of Subsection: Indication systems that integrate the display of flight or powerplant parameters to operate the airplane or are required by the operating rules of this chapter must:
  - (1) Not inhibit the primary display of flight or powerplant parameters needed by any flightcrew member in any normal mode of operation; and
  - (2) In combination with other systems, be designed and installed so information essential for continued safe flight and landing will be available to the flightcrew in a timely manner after any single failure or probable combination of failures.

FAA Advisory Circular 25.11B

- FAA Advisory Circular 25.11B offers the most comprehensive breakdown of Design, Installation, Integration, and Approval of Electronic Flight Deck displays, components, and systems installed in aircraft.
- EFIS Requires:
  - Primary Flight Display
    - Attitude
    - Altitude
    - Airspeed
    - Magnetic Heading

Garmin's Definitions within 14 CFR

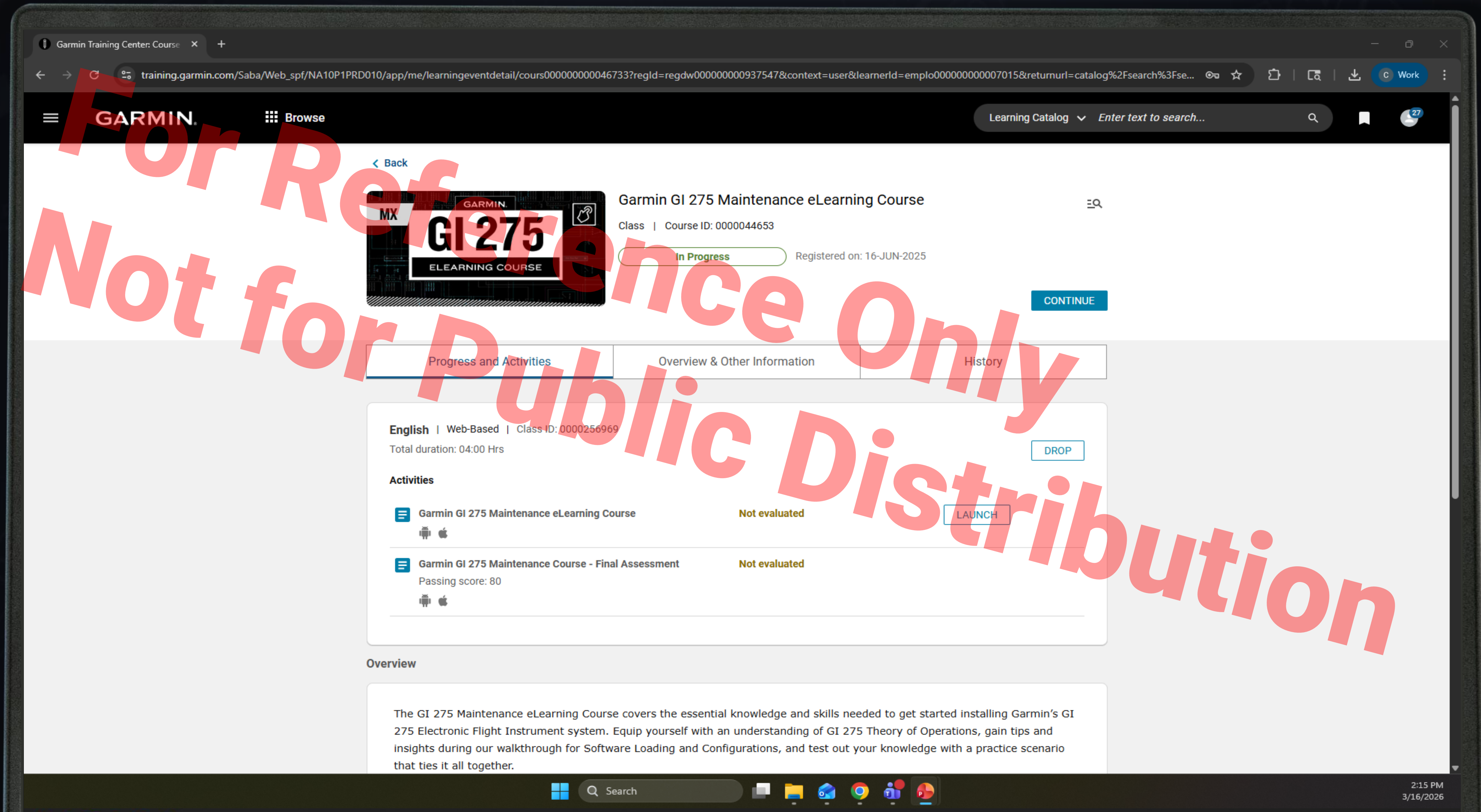
- EFIS Requires:
  - Primary Flight Display
    - Attitude
    - Altitude
    - Airspeed
    - Stabilized Heading
  - Multifunction Display
    - GPS Moving Map
    - Pilot Situational Awareness Tools
      - Ancillary and Safety of Flight
  - Engine Indication Systems

GI 275 as an EFIS System

- Single Engine Piston Aircraft:
  - PFD (ADI)
  - HSI or EHSI
  - EIS
- Minimum of 3 GI-275's
  - Base Model – Engine Indication System
  - ADAHRS Models – Primary Flight and Horizontal Situation Indicators
- Additional MFD Optional

GARMIN
AVIATION TRAINING

Aviation Training Academy



GI – Configuration Checklist

- Best workflow for EFIS Configs:
  - Identify
  - Configure
  - Calibrate

GI – Configuration Checklist

- Best workflow for EFIS Configs:
  - Identify: Upon initial start up, first we must establish the UNIT ID.
    - Required immediate Restart
    - Best to do all displays in succession at once.

GI – Configuration Checklist

- Best workflow for EFIS Configs:
  - Identify: Upon initial start up, first we must establish the UNIT ID.
    - Required immediate Restart
    - Best to do all displays in succession at once.

CAUTION

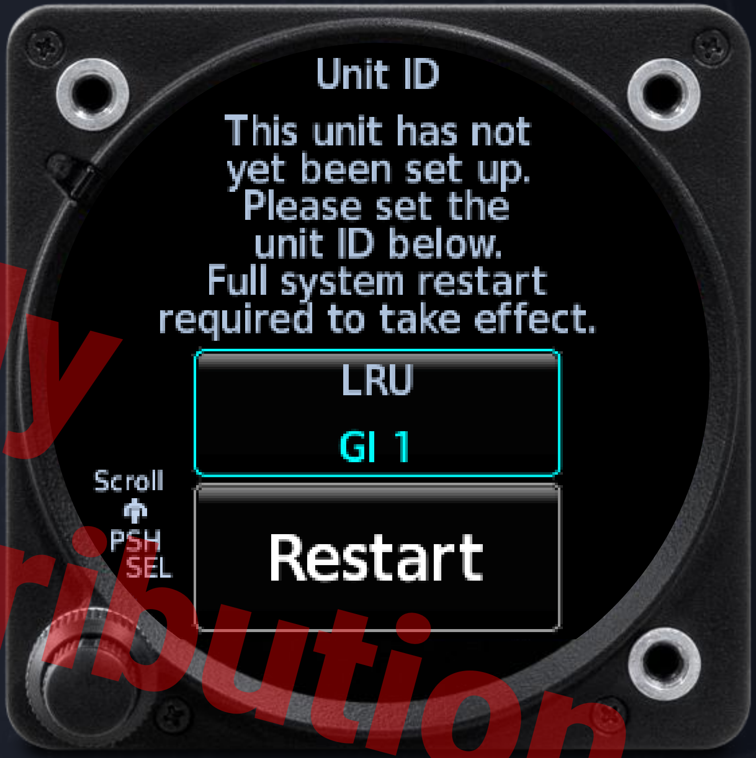
Failure to follow the procedure to set Unit ID before performing any other configuration steps may result in configuration errors or configuration settings being overwritten by another display. Unit IDs must be unique for each unit in the system.

NOTES

When the GI 275 is powered on in Configuration mode for the first time (or after configuration settings are cleared), it will automatically prompt the user to assign a Unit ID. This cannot be skipped.

GI – Configuration Checklist

- Verify all GI 275 units are powered off.
  - Power up a each GI 275 in Configuration mode per Section 5.1.1

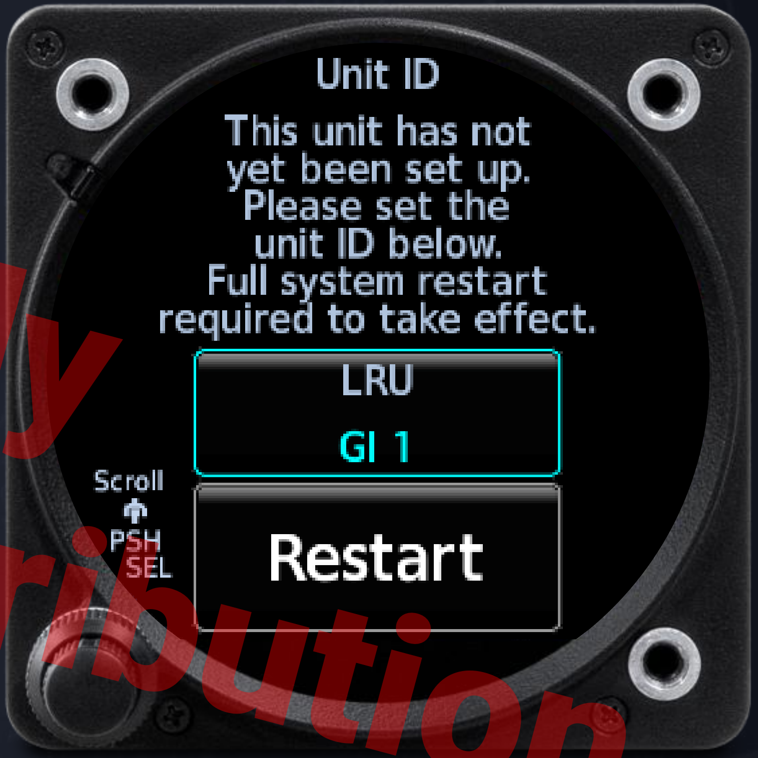


GI – Configuration Checklist

- Verify all GI 275 units are powered off.
  - Power up each GI 275 in Configuration mode per Section 5.1.1
  - Touch Unit Type → Unit ID → LRU and select a unique Unit ID between 1 and 6

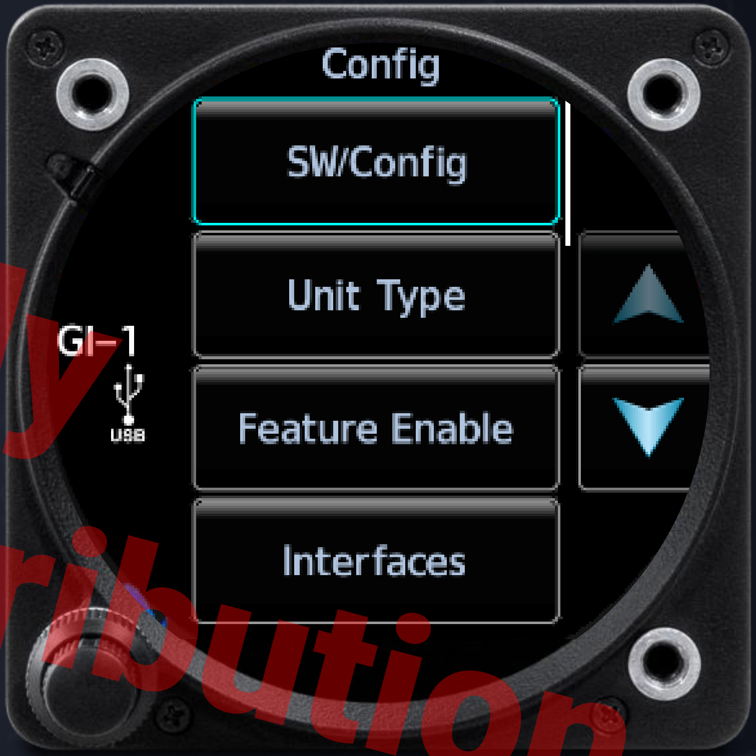
GI – Configuration Checklist

- Verify all GI 275 units are powered off.
  - Power up each GI 275 in Configuration mode per Section 5.1.1
  - Touch Unit Type → Unit ID → LRU and select a unique Unit ID between 1 and 6
  - Touch Restart to apply the assignment
    - When all Unit ID assignments have been made, power up all displays in Configuration mode before moving to the next configuration steps.



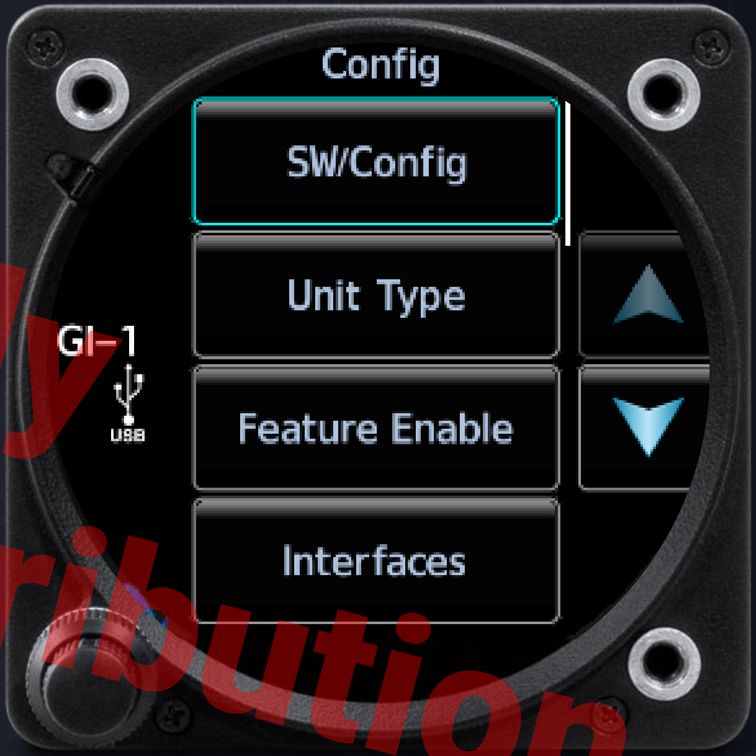
GI – Configuration Checklist

- Verify all GI 275 units are powered off.
  - Power up each GI 275 in Configuration mode per Section 5.1.1
  - Touch Unit Type → Unit ID → LRU and select a unique Unit ID between 1 and 6
  - Touch Restart to apply the assignment
    - When all Unit ID assignments have been made, power up all displays in Configuration mode before moving to the next configuration steps.



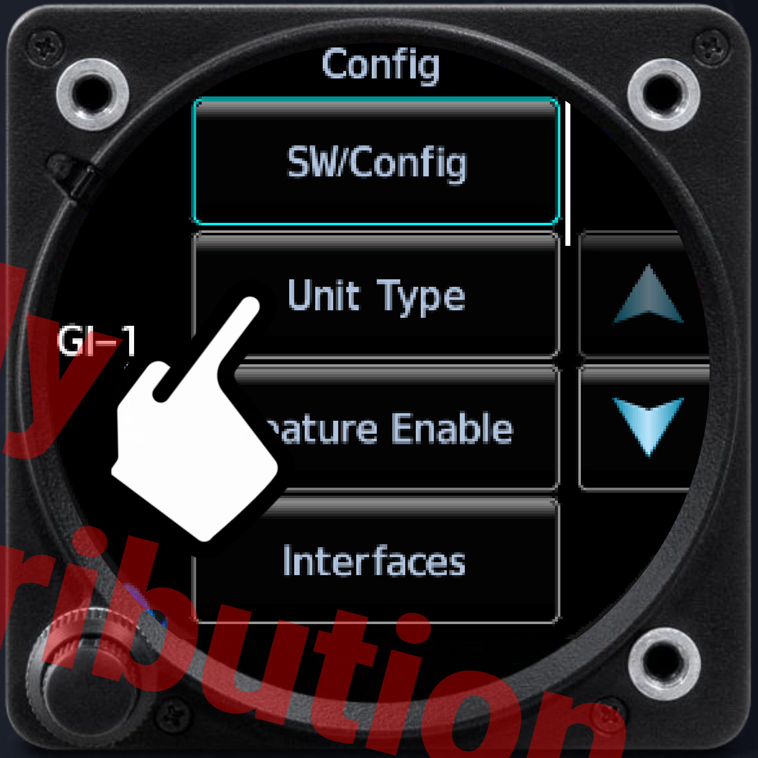
GI – Configuration Checklist

- Now we're ready to Identify Unit Type!



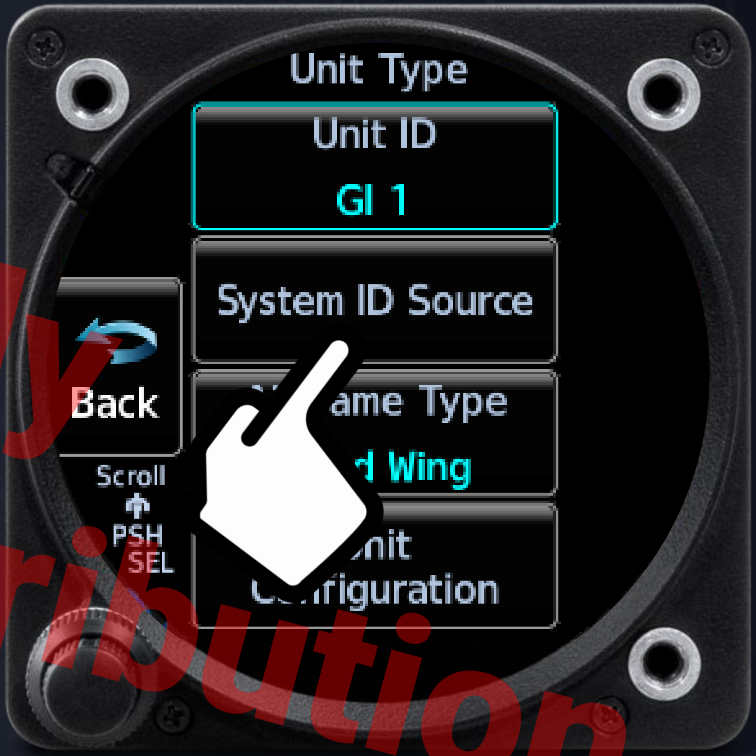
GI – Configuration Checklist

- Now we're ready to Identify Unit Type!
  - Select 'Unit Type' on the screen.



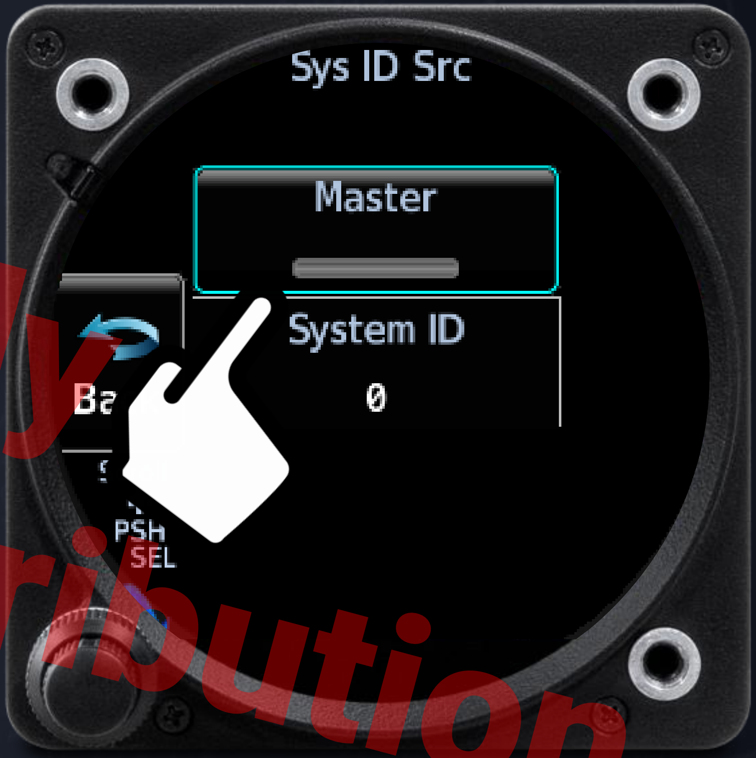
GI – Configuration Checklist

- Now we're ready to Identify Unit Type!
  - Select 'Unit Type' on the screen.
  - It's time to set the 'Master Display' for the system.



GI – Configuration Checklist

- Now we're ready to Identify Unit Type!
  - Select 'Unit Type' on the screen.
  - It's time to set the 'Master Display' for the system.
    - Notice, no 'Master Display' equals no System ID
    - We're setting the System ID for the while EFIS as a group of GI 275's



GI – Configuration Checklist

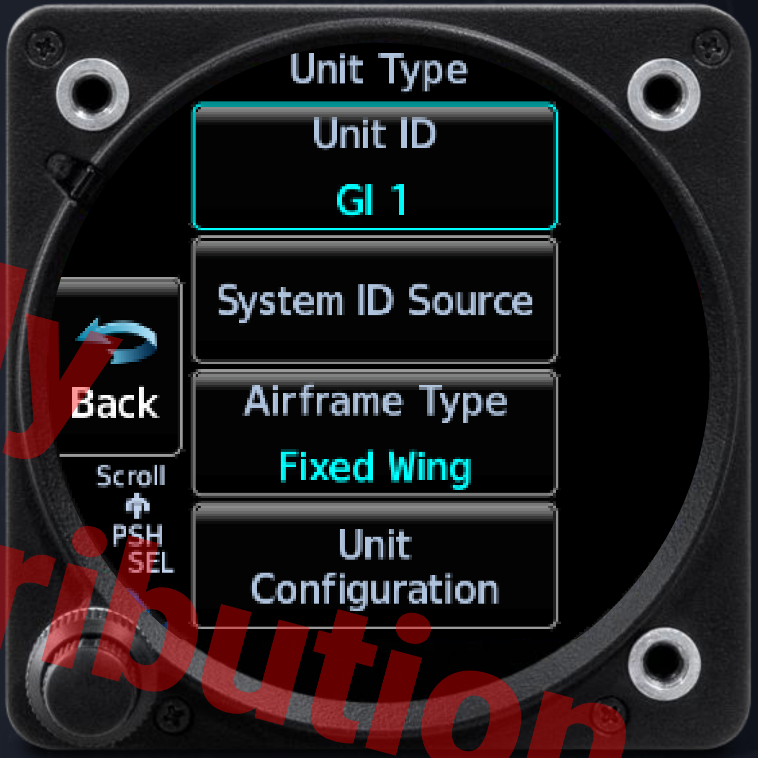
- Now we're ready to Identify Unit Type!
  - Select 'Unit Type' on the screen.
  - It's time to set the 'Master Display' for the system.
    - Notice, no 'Master Display' equals no System ID
    - We're setting the System ID for the while EFIS as a group of GI 275's
    - This will affect Enablements!

GI – Configuration Checklist

- Now we're ready to Identify Unit Type!
  - Select 'Unit Type' on the screen.
  - It's time to set the 'Master Display' for the system.
    - Notice, no 'Master Display' equals no System ID
    - We're setting the System ID for the while EFIS as a group of GI 275's
    - This will affect Enablements!

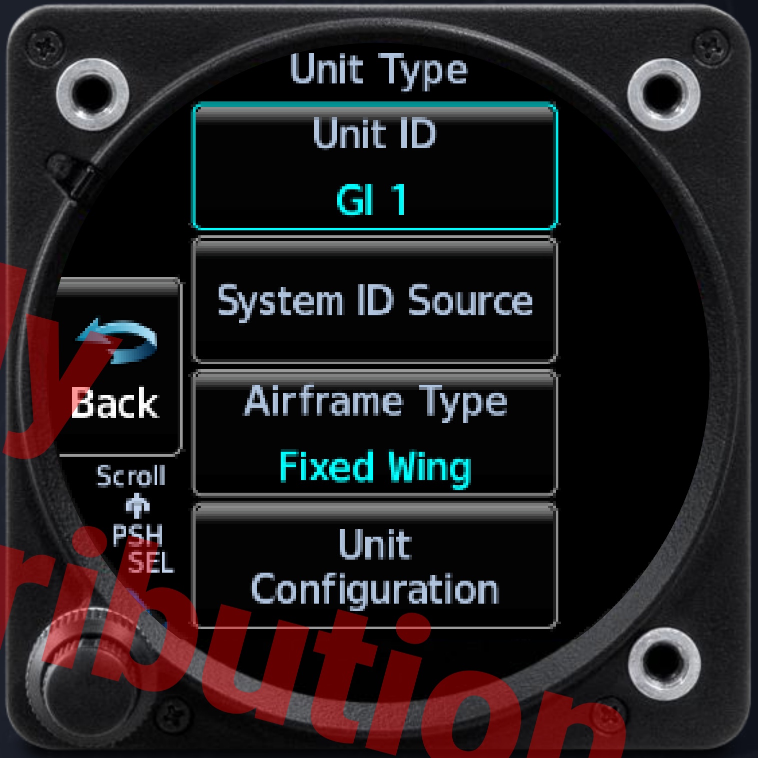
GI – Configuration

- We've established the GI's Unit ID's
- From this point forward each display must stay in Config Mode.
  - Next we'll establish the jobs for the various displays in the EFIS.
    - PFD (ADI)
    - HSI
    - EIS
    - MFD?
      - One? Two?
    - Multi-Engine EIS?
- Maximum of 6 displays in a network



GI – Configuration

- Let's establish our EFIS displays by assigning the Unit Configuration.
  - Note: The GI STC we're walking thru is the Part 23 AML STC Installation Documentation, this means we're working with 'Fixed Wing' Aircraft Type.
    - If you're working with a rotorcraft, follow the Part 27 Installation documentation.



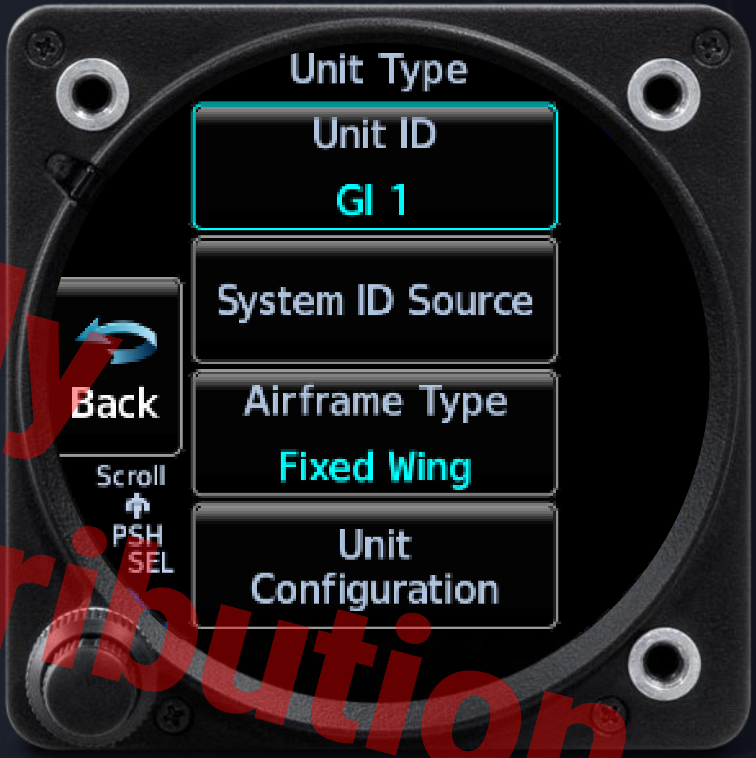
NOTES

The GI STC we're walking thru is the Part 23 AML STC Installation Documentation, this means we're working with 'Fixed Wing' Aircraft Type.

If you're working with a rotorcraft, follow the Part 27 Installation documentation

GI – Configuration

- Let's establish our EFIS displays by assigning the Unit Configuration
  - Tap: Unit Configuration

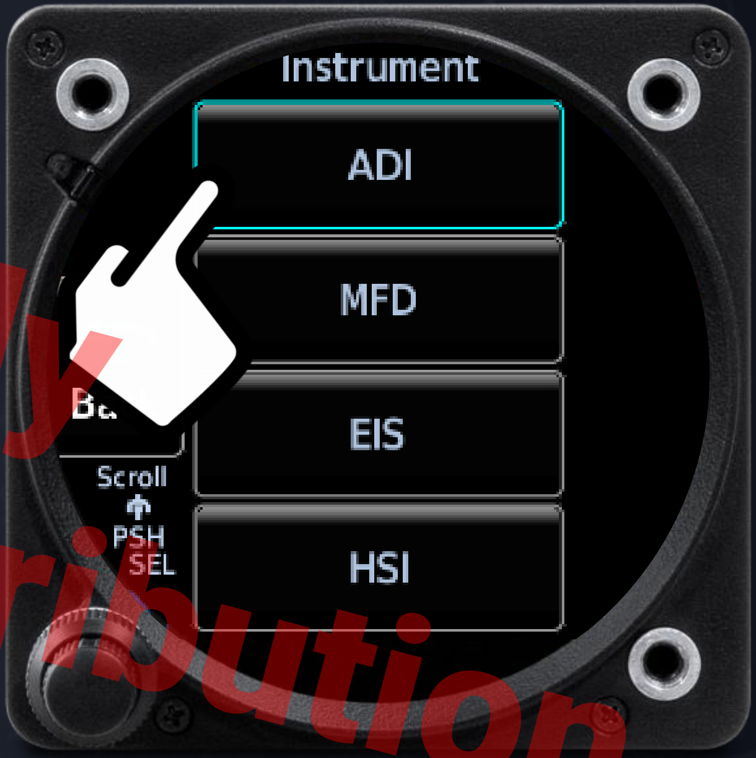


GI – Configuration

- Let's establish our EFIS displays by assigning the Unit Configuration
  - Tap: Unit Configuration
  - Tap: Instrument Type

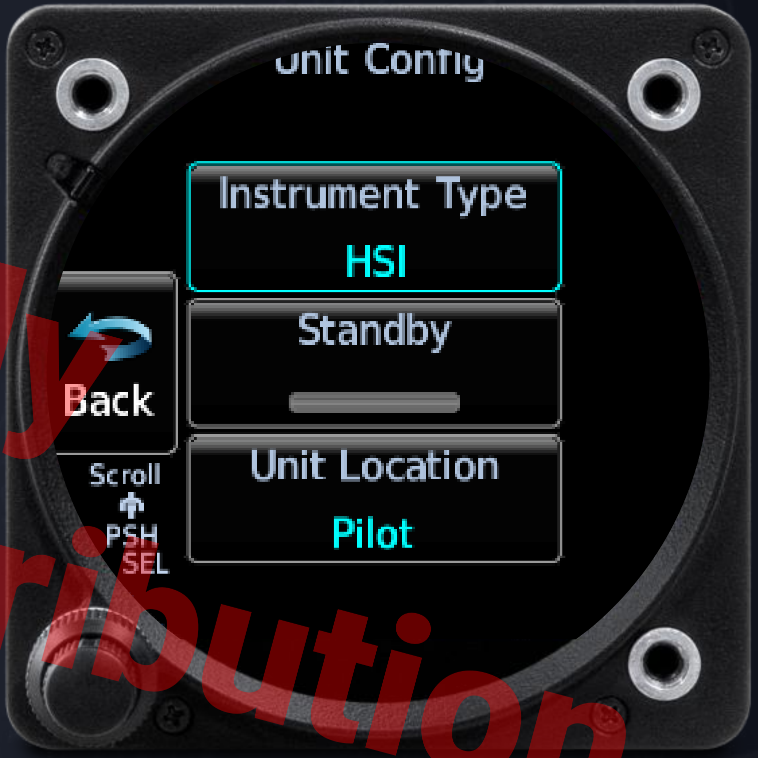
GI – Configuration

- Let's establish our EFIS displays by assigning the Unit Configuration
  - Tap: Unit Configuration
  - Tap: Instrument Type
    - Choose your Type:
      - ADI for our 'PFD'
      - HSI
      - EIS
      - MFD



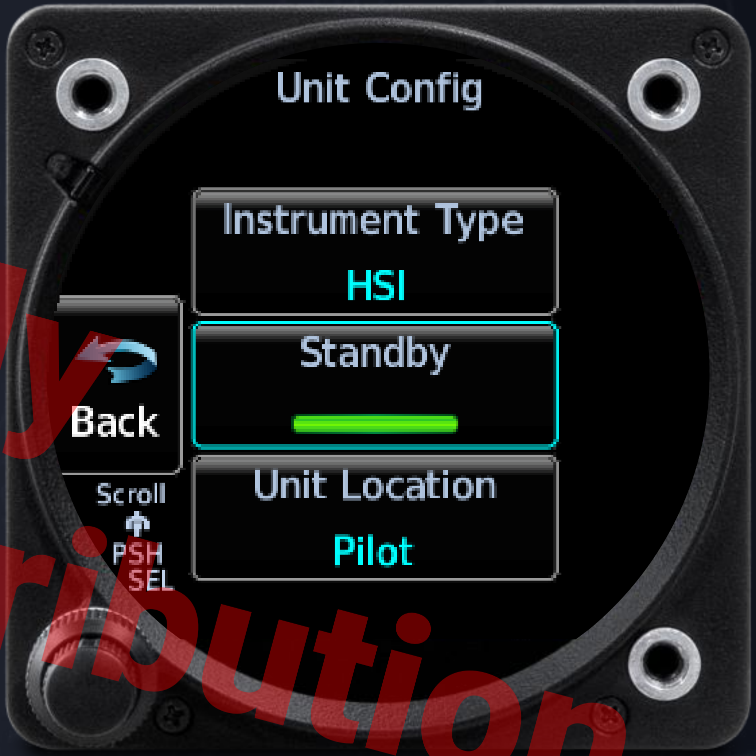
GI – Configuration

- Once we have our ADI established
  - Preferably GI 1
  - Preferably the "Smartest" GI in the system
    - Must be an ADAHRS model for ADI
- We'll want an HSI for our directional indicator or EHSI to give us our GPS Moving Map



GI – Configuration

- Once we have our ADI established
  - Preferably GI 1
  - Preferably the "Smartest" GI in the system
    - Must be an ADAHRS model for ADI
- We'll want an HSI for our directional indicator or EHSI to give us our GPS Moving Map
  - Requires ADAHRS for EHSI functions
  - Must be ADAHRS if it's to be the Integrated Electronic Standby Instrument



GI – Configuration Checklist

- Next Steps:
  - Identify
  - **Configure**
  - Calibrate

GARMIN AVIATION TRAINING

GI – Configuration Checklist

- Configure
  - Establish Interfaces
    - IFR Navigators
    - Transponders
    - Additional Systems
  - Identify LRU's for EFIS Systems
    - GMU
    - GTP
    - GEA

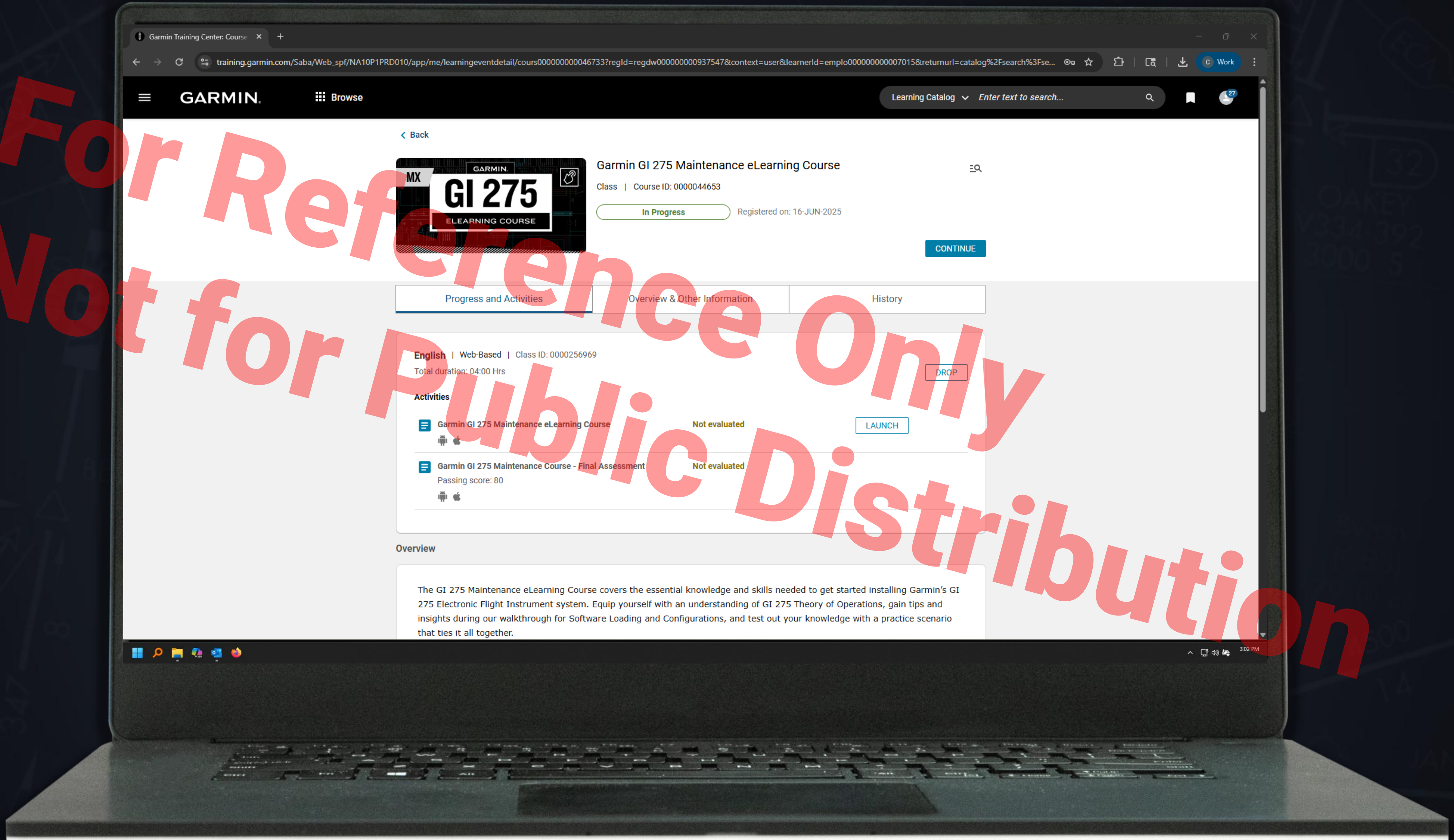
GI – Configuration Checklist

- Next Steps:
  - Identify
  - Configure
  - Calibrate

GI – Configuration Checklist

- Calibrate
  - Systems will need individual calibration for final checkout and installation delivery
  - GI 275 Part 23 AML STC Installation Manual
    - Section 5.8 covers Calibration and various system checks.

Shameless Plug



The first step for configuring the GI 275 EFIS system is:

A. Assign the System ID
B. Choosing the proper Unit ID
C. Choosing the proper Unit Type
D. Selecting the Master Display

The first step for configuring the GI 275 EFIS system is:

A. Assign the System ID
B. Choosing the proper Unit ID
C. Choosing the proper Unit Type
D. Selecting the Master Display

True or False: GI 275 offers full EFIS Function with as little as three instruments?

A. True

B. False

True or False: GI 275 offers full EFIS Function with as little as three instruments?

A. True

B. False

True or False: FAA AC 25-11b offers guidance on EFIS systems and their requirements

A. True

B. False

True or False: FAA AC 25-11b offers guidance on EFIS systems and their requirements

A. True

B. False

True or False: a pair of ADAHRS GI 275's can replace a full 6-pack of traditional analog instruments?

A. True

B. False

True or False: a pair of ADAHRS GI 275's can replace a full 6-pack of traditional analog instruments?

A. True

B. False

I participated in the Red Room EFIS Configuration Section

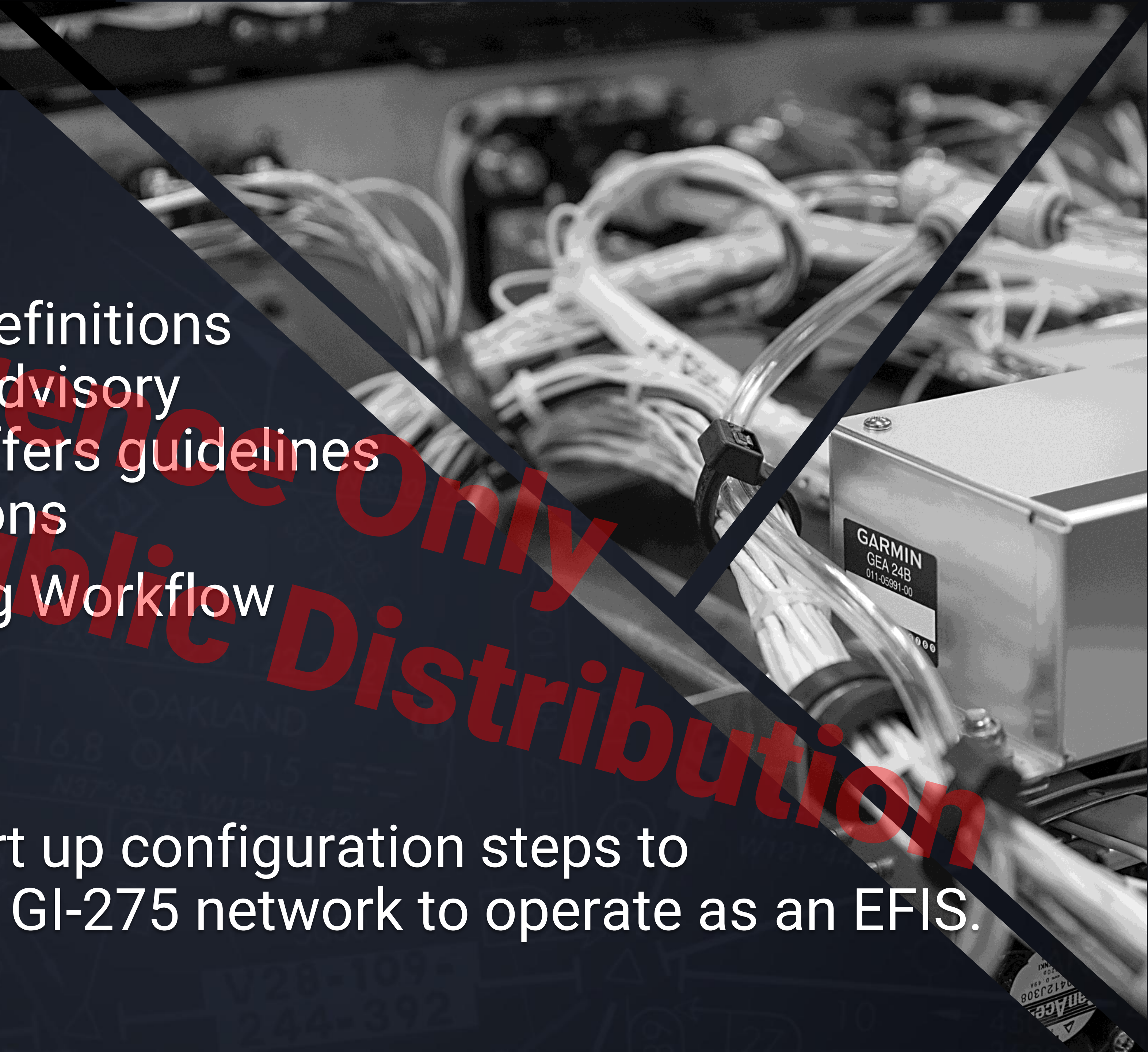
A. True

B. False

Recap

Monday, March 30, 2026

- We examined the FAA's definitions for EFIS system and the Advisory Circular (25-11B) which offers guidelines on systems and installations
- We broke down the Config Workflow
  - Identify
  - Configure
  - Calibrate
- Walked thru the initial start up configuration steps to Identify and Configure the GI-275 network to operate as an EFIS.



Any Questions?

Thank You!

- Chris Hendrix

- Senior Aviation Maintenance Instructor
- Garmin Aviation Training
- Aviation.training@garmin.com